ESAME (19/06/2025)

1) $f(x) = \frac{1}{x}\sqrt[3]{x+1}$

- $\text{Dom} f = \mathbb{R} \setminus \{0\}$ PERCHÉ $\frac{1}{x}$ É DEFINITA SOLO SE $x \neq 0$

$$= (-\infty, 0) \cup (0, +\infty)$$

- $\lim_{x \to -\infty} f(x) = \frac{-\infty}{-\infty}$ F.I.

- $\lim_{x \to -\infty} \frac{1}{x}\sqrt{x\left(1+\underbrace{\frac{1}{x}}_{\to 0}\right)} = \lim_{x \to -\infty} \underbrace{\frac{x^{\frac{1}{3}}}{x^{1}}} \cdot \left(1+\frac{1}{x}\right)^{\frac{1}{3}}$

$$x^{\frac{1}{3}-1} = x^{-\frac{2}{3}} = \left(x^{\frac{2}{3}}\right)^{-1} = \frac{1}{x^{\frac{2}{3}}}$$

- $\lim_{x \to -\infty} \frac{1}{x^{\frac{2}{3}}} \underbrace{\left(1+\frac{1}{x}\right)^{\frac{1}{3}}}_{\to 1} = \frac{1}{+\infty} = 0^{+}$

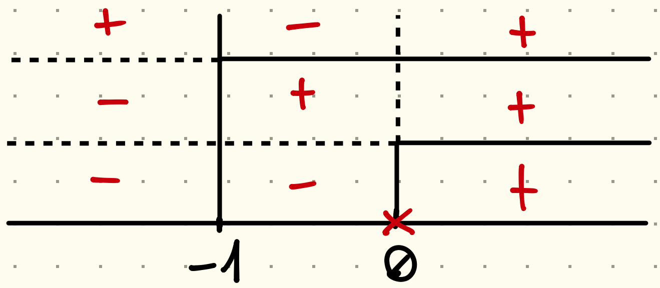
- $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} \underbrace{\frac{1}{x}}_{\to -\infty} \underbrace{\sqrt[3]{1+x}}_{1^-} = -\infty$ (con $x \to 0^-$)

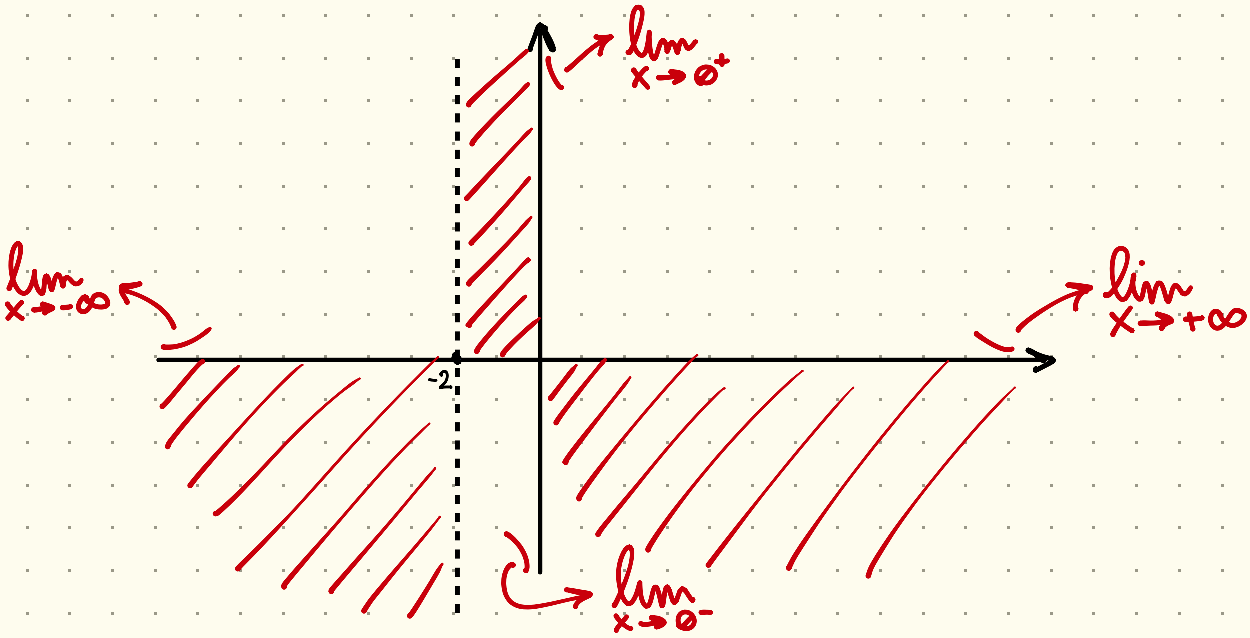
- $\lim_{x\to 0^+} \underbrace{\frac{1}{x}}_{+\infty}\underbrace{\sqrt[3]{1+x}}_{1^+} = +\infty$

- ASINTOTO VERTICALE É $x=0$ IN $x=0$
- ASINTOTO ORIZZONTALE É $y=0$ IN $y=0$

- SEGNO: $\frac{1}{x}\sqrt[3]{1+x} > 0$

  - $\frac{1}{x} > 0$ QUANDO $x > 0$
  - $\sqrt[3]{1+x} > 0 \Rightarrow x+1 > 0 \Rightarrow x > -1$





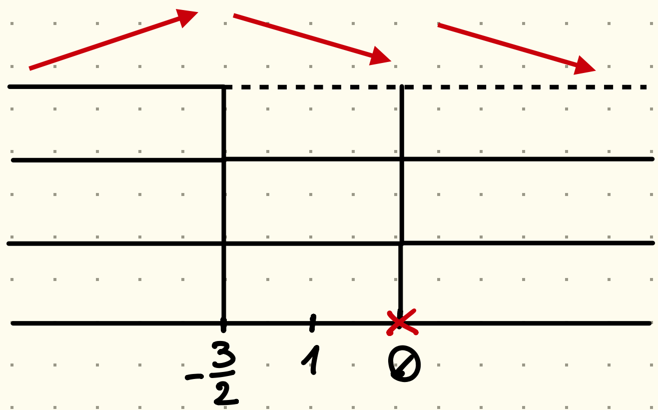
- DERIVATA

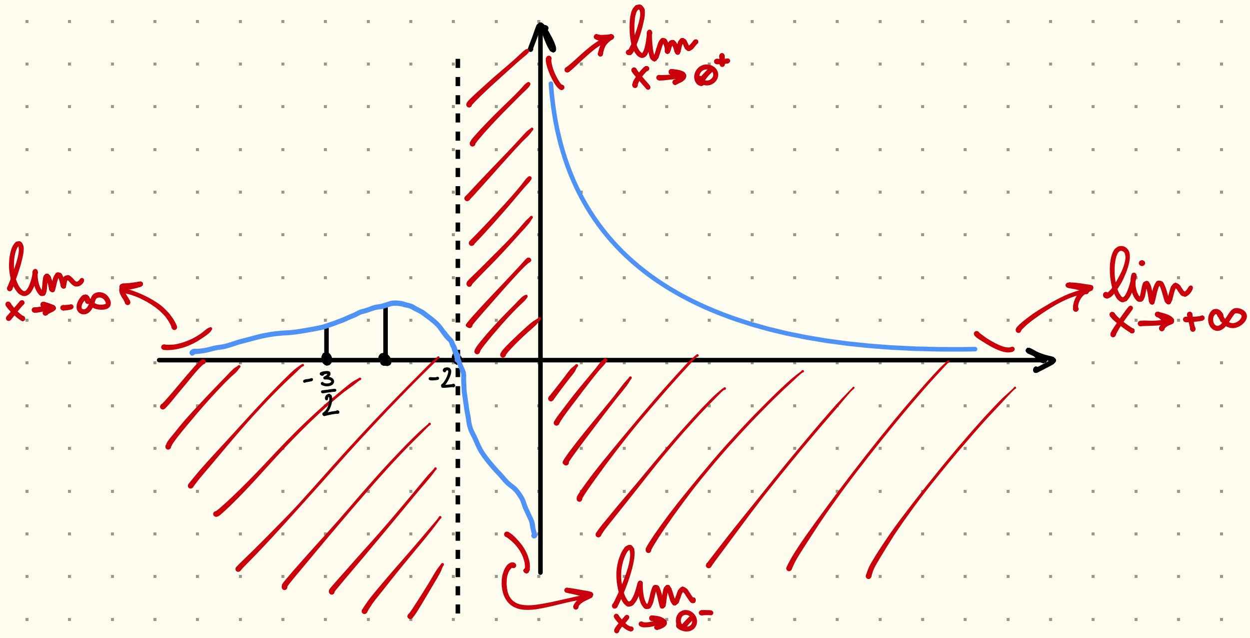
- $f(x) = \dfrac{\sqrt[3]{x+1}}{x}$

- $f'(x) = \dfrac{\frac{1}{3}(x+1)^{-\frac{2}{3}} \cdot x - 1 \cdot (x+1)^{\frac{1}{3}}}{x^2} = \dfrac{(x+1)^{-\frac{2}{3}}\left[\frac{1}{3}x - (x+1)^{\frac{2}{3}}(x+1)^{\frac{1}{3}}\right]}{x^2}$

$$= \frac{(x+1)^{-\frac{2}{3}}\left[\frac{1}{3}x - (x+1)\right]}{x^2}$$

$$= \frac{1}{(x+1)^{\frac{2}{3}}} \cdot \frac{-\frac{2}{3}x - 1}{x^2} = -\frac{\overbrace{1+\frac{2}{3}x}^{0}}{\underbrace{x^2}_{0<}\,\underbrace{(x+1)^{\frac{2}{3}}}_{>0}}$$





- f INVERTIBILE IN $x=1$?
- f DECRESCENTE IN $\left(-\frac{3}{2}, 0\right)$ → f MONOTONA → INIETTIVA
  → f INVERTIBILE